

LANDSCAPES OF THE CHURCH

01 — Predominant Land Cover

Diocesan and Episcopal Conference layers: overview, ArcGIS Online-ready descriptions, methodology, QC, formulas, workflow, and reproducible code

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6. Analysis 01 — Predominant Land Cover

6.1 Diocesan layer overview

Layer name	LandscapesChurch_01_Dioceses_Predominant_V2
Layer alias	01 - Diocesan Predominant Land Cover
Scale / features	Diocese; 2,874 features
Description	Identifies the leading and second land-cover types in each diocese and quantifies how strongly the leading type dominates the overall mapped composition.
Attribution	Catholic ecclesiastical geography from the project's versioned diocesan / Episcopal Conference framework; ecclesiastical statistics and leadership enrichment from Holy See statistical publications and Catholic-Hierarchy.org. Environmental land-cover summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons.
Citation / credits	Burhans et al., Diocesan Boundaries of the Catholic Church [Feature Layer], v1.0 and subsequent v2.0 project revision; Holy See, Annuario Pontificio / Annuario Statisticum Ecclesiae (multiple years); Catholic-Hierarchy.org (accessed Sep 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; 2026 release / project-accessed version).
Analyses	Leading/second class; leading/second share and area; dominance margin; share class.
Key story fields	Predominant-landscape, Church, scale, and contextual story fields retained in the final derivative.
Status	Publication-ready

6.2 Episcopal Conference layer overview

Layer name	LandscapesChurch_01_Conferences_Predominant_V2
Layer alias	01 - Episcopal Conference Predominant Land Cover
Scale / features	Episcopal Conference; 122 features
Description	Identifies the leading and second land-cover types across each Episcopal Conference geography and provides a concise national/multinational landscape character.
Attribution	Catholic ecclesiastical geography from the project's versioned diocesan / Episcopal Conference framework; ecclesiastical statistics and leadership enrichment from Holy See statistical publications and Catholic-Hierarchy.org. Environmental land-cover summaries derived from Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagons.
Citation / credits	Burhans et al., Diocesan Boundaries of the Catholic Church [Feature Layer], v1.0 and subsequent v2.0 project revision; Holy See, Annuario Pontificio / Annuario Statisticum Ecclesiae (multiple years); Catholic-Hierarchy.org (accessed Sep 2026); Esri, Global Biodiversity and Conservation Hexagons (Living Atlas; 2026 release / project-accessed version).
Analyses	Leading/second class; leading/second share and area; dominance margin; Conference/Church/LSAP narrative context.

Key story fields	Conference landscape and Church/LSAP story fields.
Status	Publication-ready

6.3 Final derived fields and logic

- PREDOM_LC — leading land-cover type.
- PREDOM_PCT — leading share of the analysis geography.
- PREDOM_AREA — leading-class area in km².
- SECOND_LC / SECOND_PCT / SECOND_AREA — second-ranked land-cover type and magnitude.
- PREDOM_MARGIN — difference between leading and second shares.
- PREDOM_SHARE / PREDOM_CLASS — descriptive dominance context used for interpretation.

6.4 Workflow

1. Read the eight valid land-cover percentages.
2. Sort classes descending for each feature.
3. Write leading and second class, percentage, and area.
4. Calculate leading–second margin.
5. Assign the descriptive predominance context.
6. Generate Church/Conference stories without treating the predominant class as an ecological quality judgment.
7. Run completeness and range checks; clean publication schema.

Retained QC output

```
FINAL SCHEMA / FEATURE CHECK
Dioceses:                2,874 features
Episcopal Conferences: 122 features
Final diocesan user fields: 54
Final Conference user fields: 196
Leading/second fields retained in both publication derivatives.
Area values standardized to km2 before publication.
```

Relevant reproducible code

A.4 Analysis 01 — predominant land cover

```
# raw is a dict of the eight valid land-cover percentages.
```

```
ordered = sorted(
    raw.items(),
    key=lambda kv: safe(kv[1]),
    reverse=True
)
```

```
predom_lc, predom_pct = ordered[0]
second_lc, second_pct = ordered[1]
```

```
predom_margin = (
    safe(predom_pct) -
    safe(second_pct)
)
```

```
# If source class areas have already been converted to km2:
predom_area = area_by_class_km2[predom_lc]
second_area = area_by_class_km2[second_lc]
```

A.10 Safe m² → km² conversion

```
import statistics
```

```

values = []

with arcpy.da.SearchCursor(
    feature_class,
    ["ANALYSIS_AREA"]
) as cursor:

    for (value,) in cursor:
        if value is not None and value > 0:
            values.append(float(value))

median_area = statistics.median(values)

if median_area >= 1_000_000:

    with arcpy.da.UpdateCursor(
        feature_class,
        ["ANALYSIS_AREA"]
    ) as cursor:

        for row in cursor:
            if row[0] is not None:
                row[0] = float(row[0]) / 1_000_000.0
                cursor.updateRow(row)

else:
    print("Area already appears to be in km²; no conversion performed.")

```

A.11 Rank / percentile reference implementation

```

def competition_rank(values):
    """
    values = iterable of (feature_id, numeric_value)
    Rank 1 = highest.
    Ties receive the same rank; the next rank skips accordingly.
    """
    ordered = sorted(
        values,
        key=lambda x: x[1],
        reverse=True
    )

    result = {}
    previous = None
    current_rank = 0

    for position, (feature_id, value) in enumerate(ordered, start=1):
        if previous is None or value != previous:
            current_rank = position

        result[feature_id] = current_rank
        previous = value

    return result

def percentile_at_or_below(all_values, value):
    valid = [v for v in all_values if v is not None]

    if not valid:
        return None

    n = sum(v <= value for v in valid)

    return 100.0 * n / len(valid)

```

A.16 Publication cleanup pattern

```

# Keep only fields intentionally required for the publication derivative.
KEEP_FIELDS = {
    "BUFFER_ID",
    "Account_Name",
    "Country",
    "Language",
}

```

```

"Official_website",
"Sector",
"BUFFER_AREA",
"LANDSCAPE_SCALE",
# ... eight PCT_LULC fields ...
# ... coverage/mosaic/transition fields ...
# ... story fields ...
}

delete_fields = [
    f.name
    for f in arcpy.ListFields(feature_class)
    if (
        not f.required and
        f.name not in KEEP_FIELDS
    )
]

if delete_fields:
    arcpy.management.DeleteField(
        feature_class,
        delete_fields
    )

# Always run a final required-field check after deletion.

```

Sources and attribution

Ecclesiastical geography and Church statistics: LSAP Integral Ecology Atlas / project Catholic jurisdictional framework; Holy See statistical publications; Catholic-Hierarchy.org. Environmental context: Esri ArcGIS Living Atlas Global Biodiversity and Conservation Hexagons and the underlying credited source organizations.

Esri Global Biodiversity and Conservation Hexagons: <https://www.arcgis.com/home/item.html?id=f1bdbdcac3c54ad8b5b21f5b17d91e8d>

Esri 2026 update: <https://www.esri.com/arcgis-blog/products/arcgis-living-atlas/sustainable-development/global-biodiversity-and-conservation-hexagons-but-better>

Summarize Within documentation: <https://pro.arcgis.com/en/pro-app/3.6/tool-reference/analysis/summarize-within.htm>